New monotonicity experiment: BAMNET 2025

1-minute input retained. Only the monotonicity test uses five 5-minute means.

New: 310 cold pools; 4,000 wind-only; 205 temperature-only.

Previous 1-minute result: 187 / 4,117 / 148. Original 5-minute: 197 / 2,294 / 178.

205 thermal-only entries = 148 valid episodes +57 legacy missing-mask QA entries.

All 57 available files were processed. No satellite verification is performed.

Exact new monotonicity definition

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ONLY MONOTONICITY CHANGES

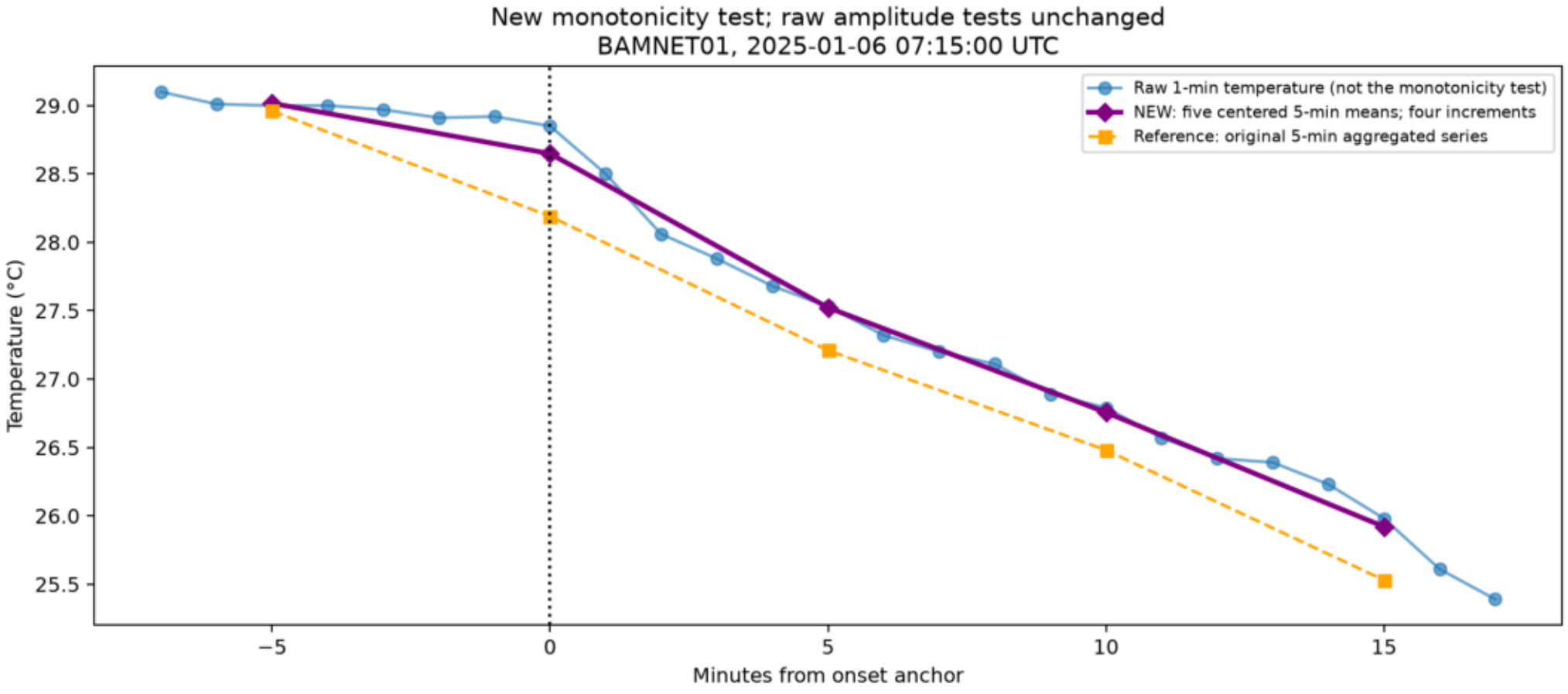
Input remains 1-minute temperature and 1-minute gust.
Centers of the five means: −5, 0, +5, +10, +15 minutes.
Each mean uses five values: center−2 .. center+2 minutes.
Compare the four consecutive differences of these five means.

Raw T(t+15)−T(t−5) <= −2 C remains unchanged.
Raw T(t)−T(t−5) <= −0.5 C remains unchanged.
Wind equations, thresholds, grouping and legacy mode are unchanged.

Important: mean support extends from t−7 to t+17 minutes.
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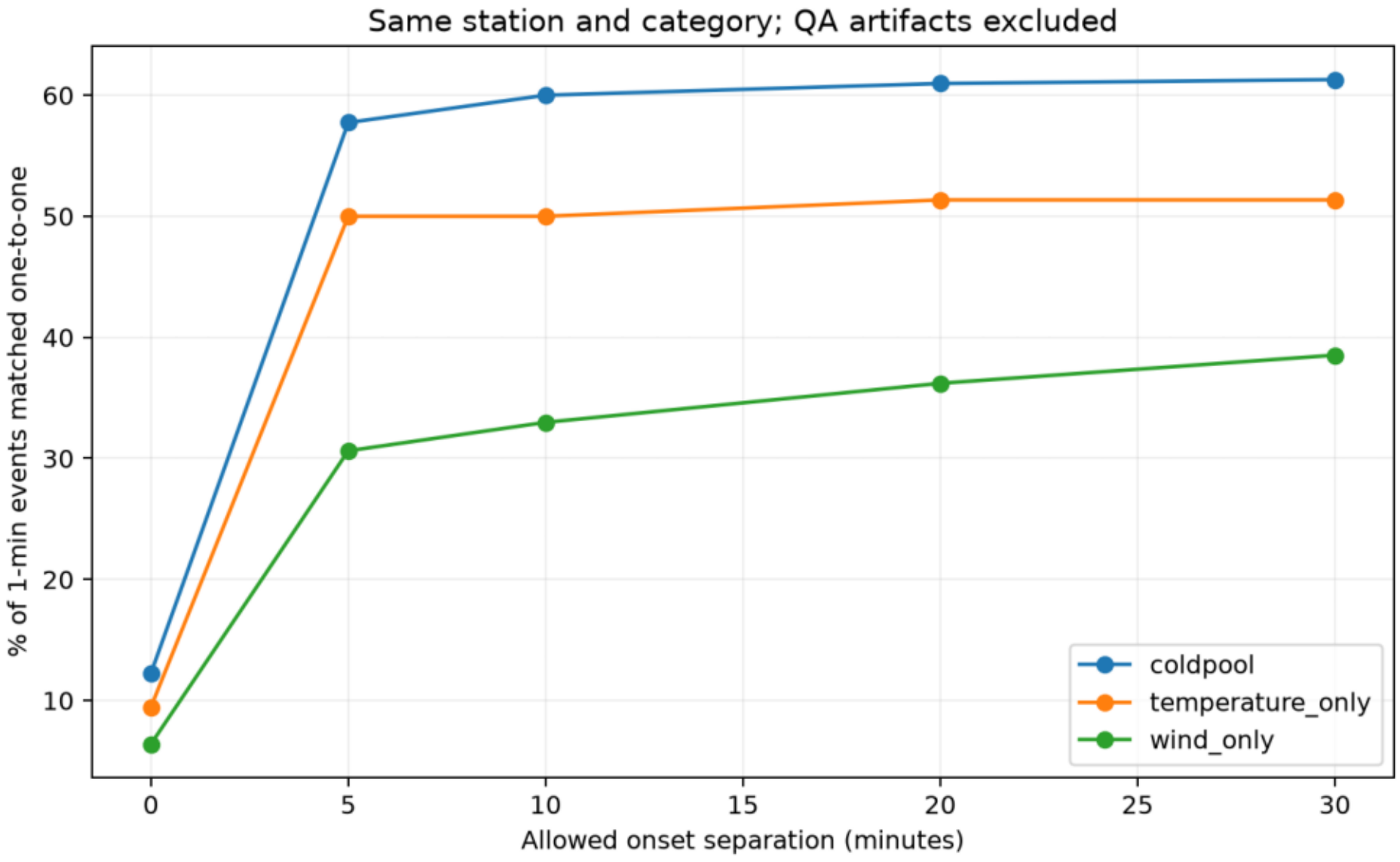
No minute-scale monotonicity test remains; raw drop amplitudes remain unchanged.

A real temperature example



Purple: five centered means from minute data; compare four increments.

Overlap with original 5-minute detections



±20 min: 189 cold-pool pairs; 61.0% of new, 95.9% of 5-minute events.
Same station, one-to-one matching; QA excluded.

Changes relative to the previous 1-minute algorithm

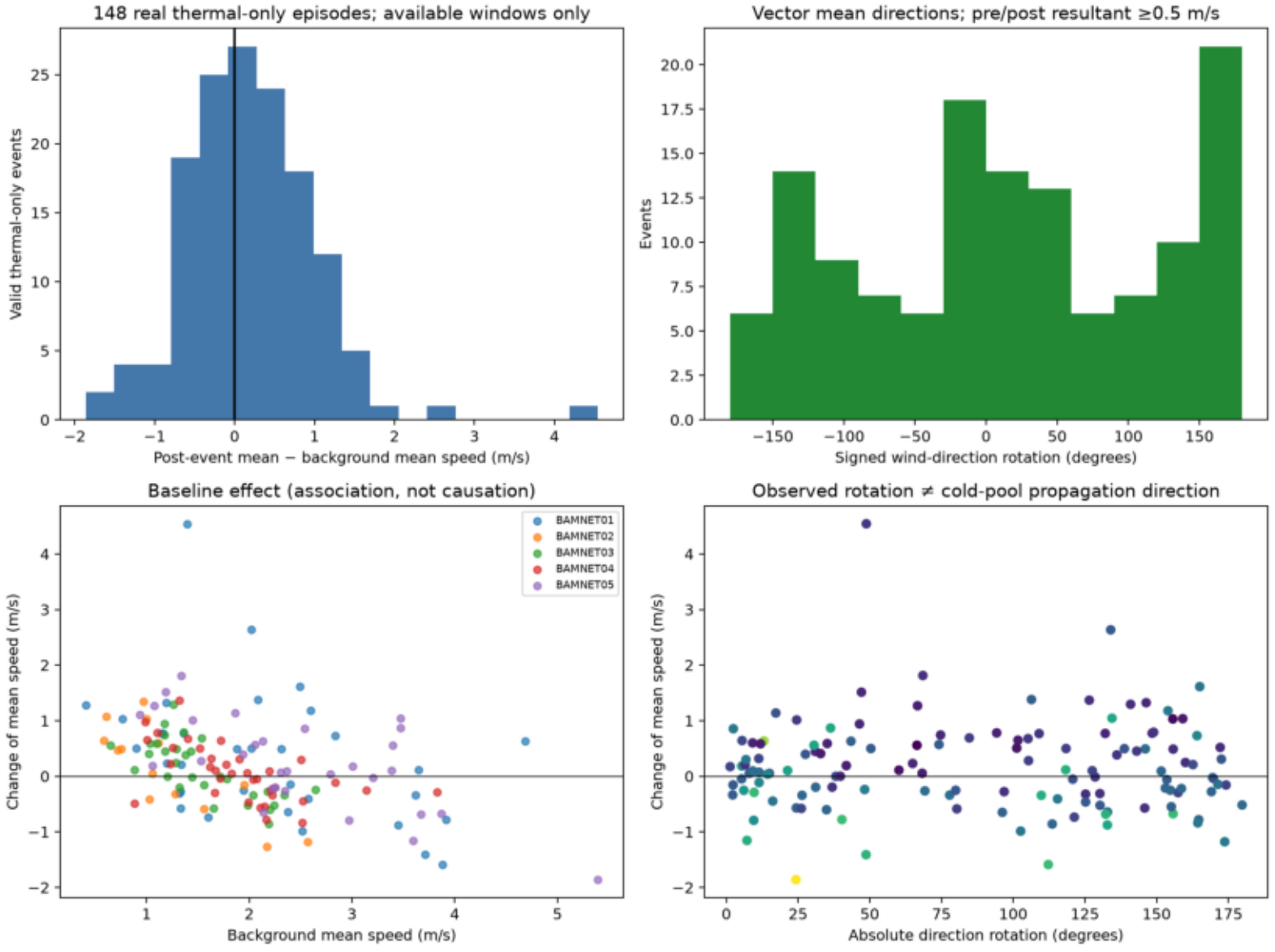
Cold-pool pairs within ±20 minutes: 187 (old 187 versus new 310).

More candidates are not automatically more accurate detections.
The modified constraint is an explicit sensitivity experiment.

Five centered means span 25 source minutes (t−7..t+17).
It is impossible to form five disjoint 5-minute blocks from only 21 samples.

Raw drop tests still use t−5..t+15 and t−5..t.
Wind equations and event grouping are unchanged.

Thermal-only wind response and baseline



Valid thermal-only: {'increase': 81, 'decrease': 61, 'missing': 5, 'unchanged': 1}.
Background −40..−20; event 0..20 min; minimum 17/21 samples.

Direction and baseline results

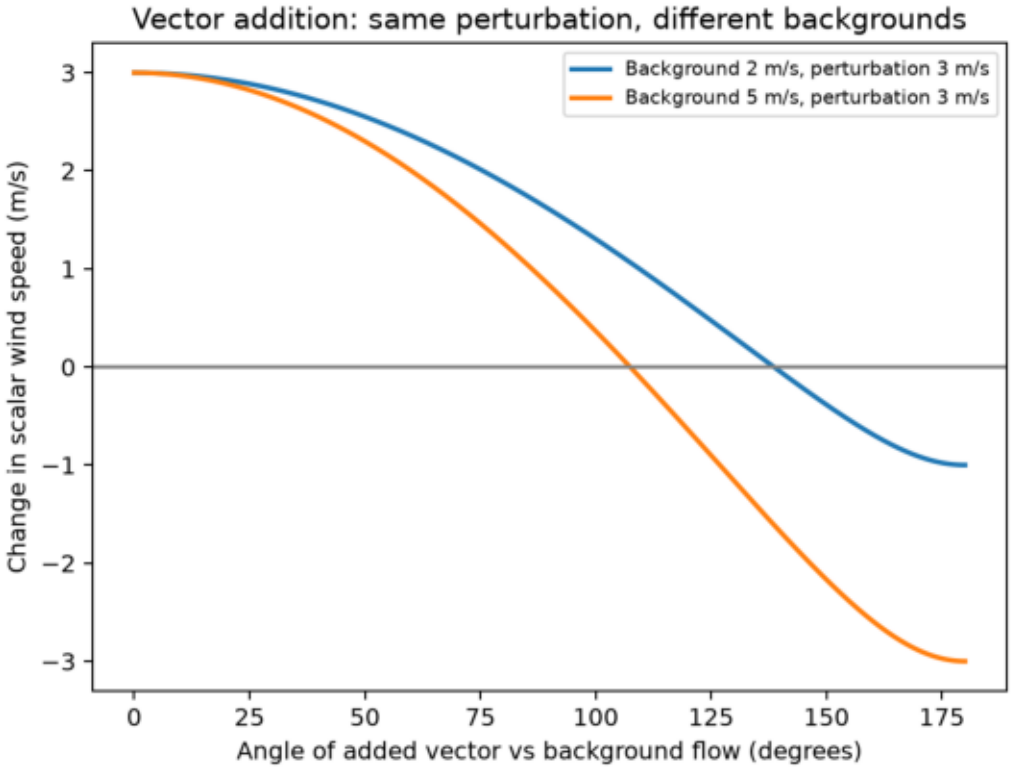
Median speed change: 0.110 m/s; n=143.

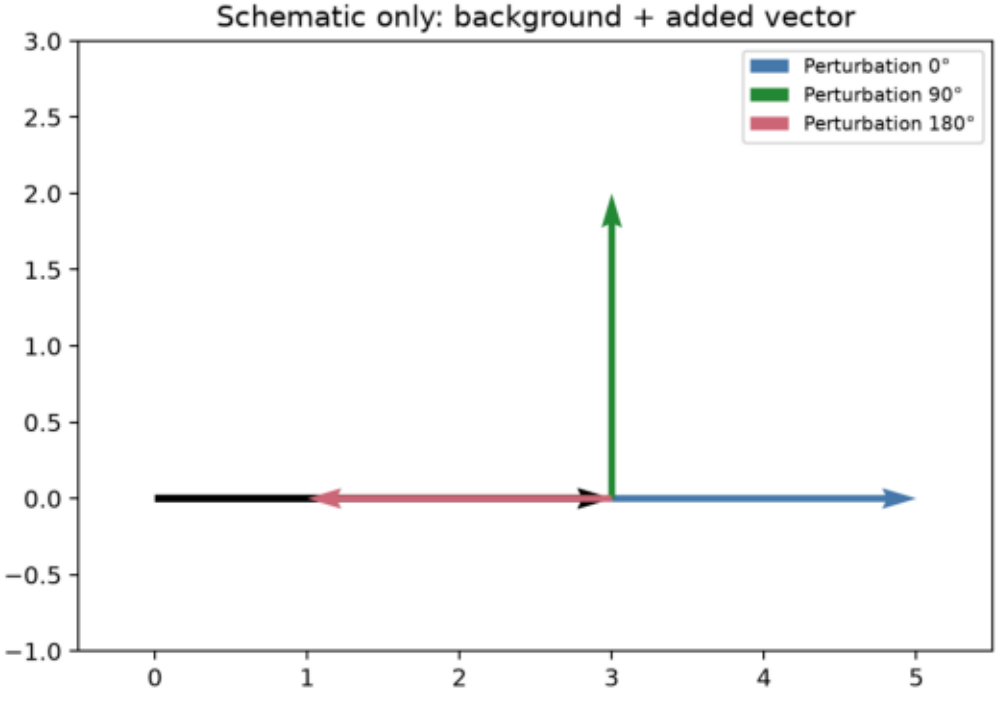
Reliable vector-mean directions: 131; median absolute rotation 96.2°.

Baseline vs mean-speed change: rho=-0.489.
Baseline vs peak change: rho=-0.117.

These are descriptive associations. Subtracting baseline creates mathematical coupling; local wind direction is not propagation direction.

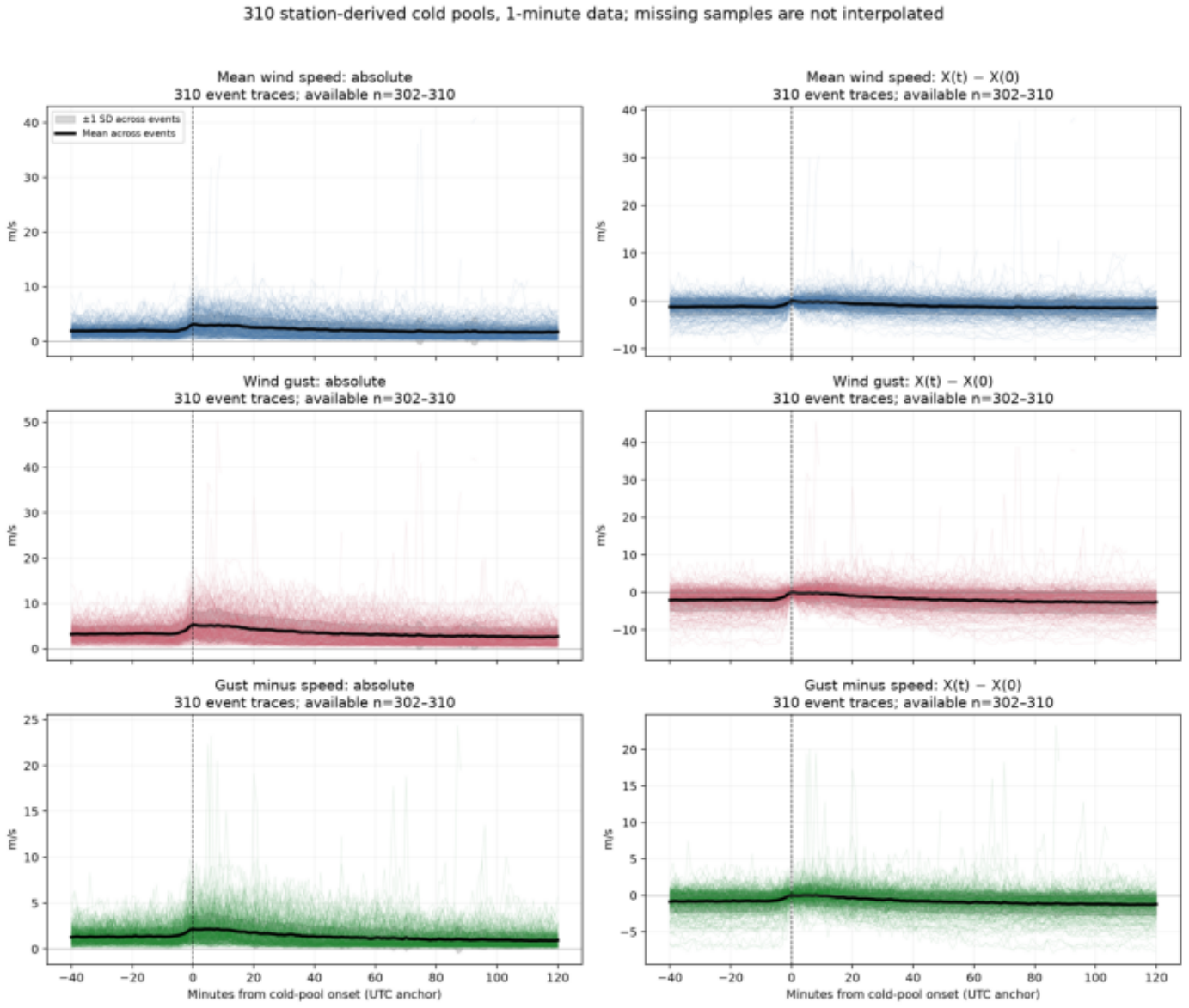
Vector geometry: interpretation limits





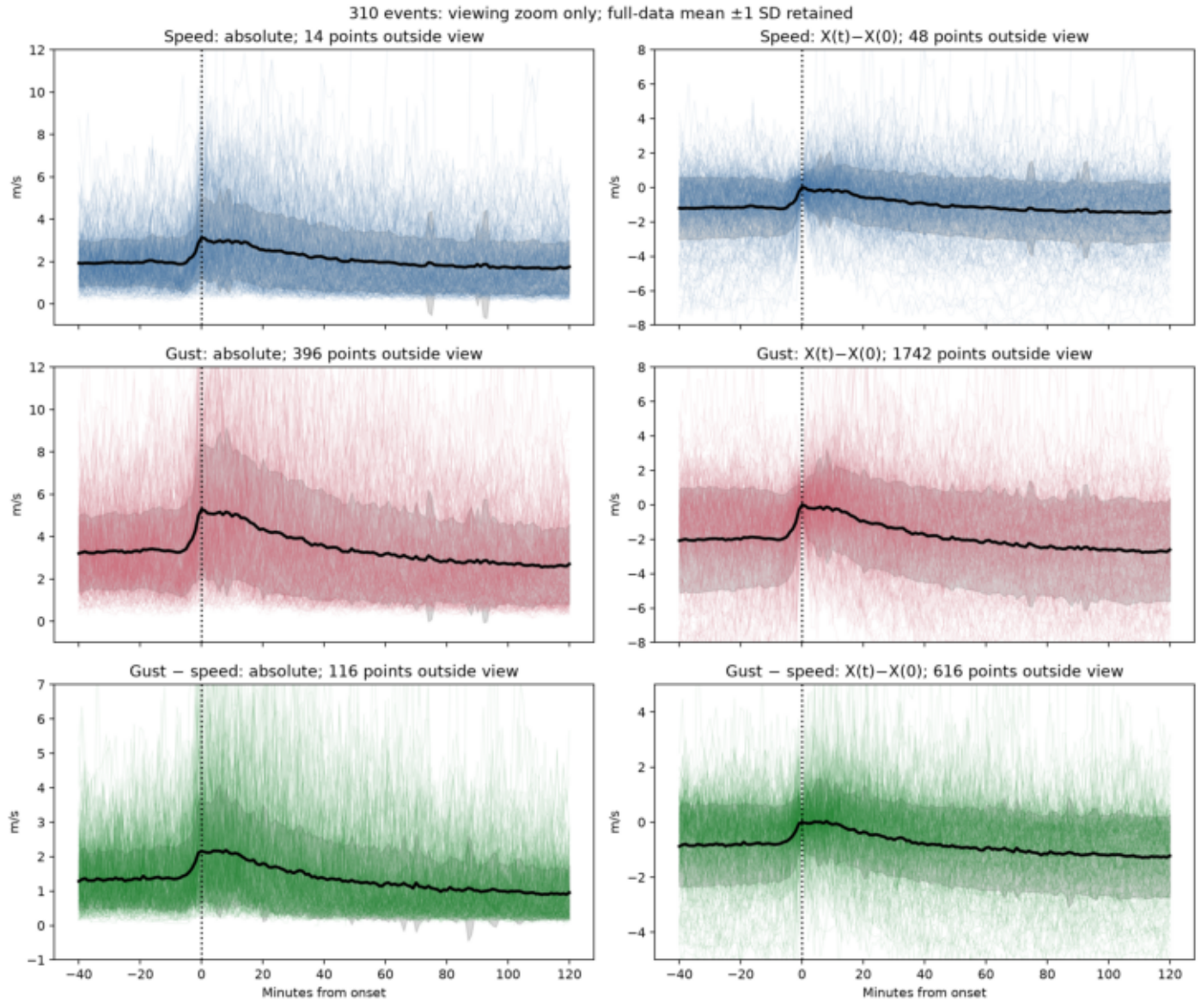
Schematic only, not inferred cold-pool motion.

Six-panel composite: 310 selected events



Thin event traces; thick black mean; shading ±1 population SD. Left absolute; right X(t)−X(0).
Window −40..+120 min. Missing values are not interpolated.

Viewing zoom, not data exclusion



All data are retained in the mean and standard deviation.

Speed versus gust and next decisions

Paired eligible events: 308. Median peak increase:
speed 2.265 m/s; gust 3.888 m/s.

Gust-selected samples cannot establish that gust has higher accuracy.
Use independent labels and the same thermal candidates for a fair comparison.

Normalization: subtract each event onset value, then calculate ensemble mean/SD.
Units remain m/s; not division, percentage change, or z-score.

Next meeting: confirm the centered averaging convention, assess QA artifacts,
validate sensitivity against independent evidence, and investigate direction.